DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	**MANAGEMENT OF PEDIATRIC DIABETIC KETOACIDOSIS**		
Department	**Pediatric Intensive Care Unit**		
Applies to	**All PICU Medical and Nursing Staff**		
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1.0 PURPOSE

1.1. Management of DKA patients in a proper way with prevention of complications.

2.0 DEFINITION/ABBREVIATION

2.1. **Diabetic Ketoacidosis (DKA)**- serious complication of diabetes that can be life-threatening. DKA develops when your body doesn't have enough insulin to allow blood sugar into your cells for use as energy. Instead, your liver breaks down fat for fuel, a process that produces acids called ketones.

2.2. Criteria for diagnosis of DKA

2.2.1. Hyperglycemia- blood sugar level of >11-13.5 mmol/L (200-250 mg/dl)

2.2.2. Metabolic acidosis- pH level of <7.30

2.2.3. Bicarbonate- <15 mmol/L

2.2.4. Glucosoria and ketonuria

2.2.5. Ketonaemia (β-OHB and acetoacetate > 3 mmol/L)

2.3. Classification of DKA

	Normal	Mild	Moderate	Severe
pH	7.35-7.45	7.25-7.34	7.15-7.24	<7.15
CO_2 venous	20-28	16-20	10-15	<10
HCO3	18-24	<15mmol/L	<10mmol/L	<5 mmol/L
Clinical	-no change	-oriented -alert -fatigued	-kaussmaul respirations -oriented -sleepy but arousable	-kaussmaul or depressed respiration -sleepy to depressed sensorium

3.0 RESPONSIBILITY

3.1 **PICU consultant/specialist** in-charge of the unit responsible for the medical care of the patients

3.2 **Head Nurse:** Responsible in ensuring that all staffs are competent in handling procedure and in monitoring the actual performance of the staff nurse.

3.3 **Staff nurse:** Responsible in carrying out doctor's order and monitoring patient's condition.

4.0 POLICY

4.1. Patients must fulfill diagnostic criteria in ER. The patient/s should be stabilized before shifting to PICU.

4.2. In PICU, applying the exact plan for fluid, electrolyte and insulin therapy.

4.3. Recognition and prompt treatment of complications (especially cerebral edema.)

4.4. In the ward: regular monitoring, RBS adjusting insulin dose, intensive health education (use of glucometer, insulin: dose calculation injection)

4.5. Before discharge, ensure good health education about TIDM, how to use glucometer, how to inject insulin, management of hypo/hyperglycemia. Appointment to diabetic clinic for regular follow up.

4.6. Follow the indications of Admissions to PICU

4.6.1. pH <7

4.6.2. Age <5 year

4.6.3. Comatose

4.6.4. Blood glucose >1000 mg/dl (55 mmol/L)

4.6.5. >10% dehydrated with shock

4.6.6. Staffing levels on the wards are insufficient to allow adequate monitoring dysrhythmia

5.0 PROCEDURE (S)

5.1. Initial assessment and resuscitation

5.1.1. Airway and breathing –rarely a concern

5.1.2. Circulation-usually 7-9% dehydrated

5.1.3. Neurological state-rapid assessment, expect improved level of consciousness with initial resuscitative measures.

5.1.4. Give oxygen even if saturation is 100% RA

5.2. Fluid therapy must be initiated within the first hour.

5.2.1. Bolus: 10-20ml/kg (NS 0.9% NaCl or Ringer Lactate)

5.2.2. Alkali therapy:

5.2.2.1. NaHCO3 should only be given if pH is <6.9 or patient is with poor clinical condition (shock, symptomatic hyperkalemia) single, small dose 1-2 meq/kg maximally (80mmol/m2) over 1-2 hours.

5.2.3. Confirm diagnosis

5.2.4. Fluid and electrolyte therapy from 2nd hour till resolution:

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5.2.4.1. ½ NS 0.45% NaCl, add K if the patient is passing urine;

5.2.4.2. Insulin drip: Add 50 units' regular insulin to 500 ml (NS 1 ml=0.1 units by separate pump, rate of infusion= 0.1 unit/kg/h

5.2.5. Identification and treatment of precipitating factors

5.2.6. Treatment of complications

5.3. Initial lab works to be started:

5.3.1. CBC and differential count

5.3.2. Serum electrolyte (Na, K, urea, creatinine, Ca, phosphate, Mg)

5.3.3. Glycated hemoglobin HbA1c

5.3.4. Serum amylase and lipase if there is abdominal pain

5.3.5. Blood and urine cultures: H/O or signs suggestive of sepsis such as fever, focus of infection, history suggestive of UTI or pneumonia)

5.3.6. Baseline ECG

5.3.7. Imaging

5.4. Monitoring of DKA

5.4.1. Clinical: vital signs every 30 minutes, coma score every 30 mins- 1 hour, ECG if k is <3 or > 6 mmol/L

5.4.2. Lab:

5.4.2.1. Serum glucose monitoring every hour til DC IV insulin

5.4.2.2. VBG every hour during severe DKA, every 2H for pH 7.1-7.19, every 4H for pH 7.2-7.29

5.4.2.3. Electrolytes every 2H during severe DKA then every 4H

5.4.2.4. Calcium, magnesium and phosphorous every 4H

5.4.2.5. Urine: ketones and glucose every void

5.5. Electrolytes

5.5.1. Potassium (K) management

5.5.1.1.	Serum K	Add to IVF
5.5.1.2.	<2.5	1 meq/K over 1 hour, don't give insulin

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5.5.1.3. 2.5-3.5 added 40-60 mEq/L

5.5.1.4. 3.5-5 added 20-30 mEq/L

5.5.1.5. 5-5.5

5.5.2. Sodium (Na) management

5.5.2.1. If initial Na is usually normal or low, estimate of true Na for any glucose >100mg/dl (5.6mmol/L)

$$\text{serum Na+} \frac{\text{glucose (mg/dL) -100}}{100} \times 1.6$$

5.5.2.2. One should expect Na to increase 1.6 mmol/L for each 100mg (5.6 mmol) glucose

5.5.2.3. If corrected Na> 150mmol/L severe hyperNa may be present; In that case, osmolality will be high and rehydration should be slow over 48 hours.

$$\text{IVF rate} = \frac{\text{85ml/kg + maintenance+ bolus}}{\text{23 hours}}$$

85 ml/kg = 8.5% deficit

Maintenance
1st 10 kg: 100ml/kg
2nd 10 kg: 50ml/kg
all remaining kg: 25 ml/kg

Example: 30 kg child

-1st 10 kg: 10x100= 1000 ml- 2nd 10 kg: 10x50=500ml
3rd 10 kg: 10x25= 250 ml- so 30 kg= 1750

$$\text{IV rate} = \frac{\text{(85x30) + 1750-300 ml (bolus 10ml/kg)}}{\text{23 r}}$$

= 175 ml/hr (including insulin infusion which is 30 ml/hr) so IV rate will be 145 ml/hr

5.5.3. Alkali therapy: if not given in ER: NaHC03 should be given if:

5.5.3.1. pH is 6.9

5.5.3.2. poor clinical condition (shock, symptomatic hyperkalemia)

5.5.3.3. single small dose 1-2 mEq/kg maximally 80 mmol/m2 over 1-2 hours

5.5.4. Management of fluid (still resolution of acidosis)

5.5.4.1. Insulin infusion should remain at or above 0.1 u/kg/hr until resolution of acidosis

5.5.4.2. If blood glucose falls <17mmol (300 mg) add D5W

5.5.4.3. Maintain blood glucose 100-250mg/dL

5.5.4.4. If the blood glucose falls <8mmol/L (increase dextrose to D10W up to a maximum of D12.5%W)

5.5.4.5. If still hypoglycemic, decrease insulin infusion by decrement of 0.02 u/kg/hr.

5.6. Management of cerebral edema

5.6.1. Contributing Factors

5.6.1.1. Determine the risk factors:

5.6.1.1.1. Children <5 yrs

5.6.1.1.2. Those with newly diagnosed DKI

5.6.1.1.3. More severe acidosis

5.6.1.1.4. More severe dehydration or increased BUN at presentation

5.6.2. Determine the iatrogenic factors:

5.6.2.1. High rate of fluids >4L/m2/24H

5.6.2.2. Bicarbonate administration

5.6.2.3. Falling serum Na or failure of serum Na to increase during treatment

5.6.3. Bedside symptoms and signs of cerebral edema

5.6.3.1. Diagnostic criteria

5.6.3.1.1. Decorticate or decerebrate postures

5.6.3.1.2. Cranial nerve palsy (3,4,7)

5.6.3.1.3. Abnormal neurogenic respiratory pattern (grunting, tachypnea, Cheyne-stockes respiration, apnea)

5.6.3.2. Major criteria

5.6.3.2.1. Altered mentation

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5.6.3.2.2. Sustained HR deceleration (20 beats/min) not attributed to improved intravascular volume or sleep rate

5.6.3.2.3. Age-inappropriate incontinence

5.6.3.3. Minor criteria

5.6.3.3.1. Vomiting

5.6.3.3.2. Headache

5.6.3.3.3. Lethargy

5.6.3.3.4. Diastolic BP > 90mmHg

5.6.3.3.5. Age younger than 5 yrs

5.6.4. **<u>One diagnostic criterion and two major criteria, OR one major and two minor criteria</u>**

5.6.4.1. Other complications: hypoglycemia, hypokalemia and persistent acidosis

5.6.4.2. Treatment of cerebral edema

5.6.4.2.1. Secure airway, intubate and ventilate if needed

5.6.4.2.2. Raise the head of the bed 30 degrees

5.6.4.2.3. IVF rate to maintenance or less

5.6.4.2.4. IV mannitol (mainstay of treatment- 0.25-0.5g/kg) over 20 mins followed by other dose after 2 hours if still needed

5.6.4.2.5. Patients may need hyperventilation but aggressive hyperventilation has been associated with poor outcomes and is not recommended. The aim is to keep C02 <30-35 mmHg

5.6.4.2.6. Neurosurgical consultation should be obtained and ICP monitoring begun

5.6.4.3. Post-resolution Phase (transition to subcutaneous insulin)

5.6.4.3.1. Total HCO3 >15 mEq/ pH>7.3

5.6.4.3.2. Serum Na stable between 135-145 mEq/L

5.6.4.3.3. No emesis

5.6.4.3.4. Transition better to be done in the morning

5.6.4.3.5. Do not give SC insulin until meal has actually been eaten

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5.6.4.3.6. Wait 30 minutes after meal; if there is no emesis, give only *regular insulin* dose SC (according to sliding scale) then wait 30 minutes then DC infusions and observe for 1-2 hours then transfer to the ward.

5.6.4.3.7. Ward management

5.6.4.3.7.1. Identification and treatment of precipitating factors

5.6.4.3.7.2. Adjustment of insulin dosage

5.6.4.3.7.3. Education of the patient and the parents

5.6.4.3.7.4. Nutritional meal

5.6.4.3.8. Adjustment of insulin dose

5.6.4.3.8.1. Check Dx-sticks q 6 hour:

5.6.4.3.8.2. -***If the patient has established diabetes,*** the previous home regimen should be followed.

5.6.4.3.8.3. -***If newly diagnosed***, use sliding scale with regular SC insulin once IV infusion stop

5.6.4.3.8.3.1. < 5 mmol/L = none

5.6.4.3.8.3.2. 5-10 mmol/L = 0.2 u/kg

5.6.4.3.8.3.3. 10-15 mmol/L = 0.3 u/kg

5.6.4.3.8.3.4. 15-20 mol/L = 0.4 u/kg

5.6.4.3.8.3.5. >20 mmol/L = 0.5 u/kg

5.6.4.3.8.3.6. Then the next day, start double regimen which is adjusted according to the age of patient

Prepubertal	**Pubertal**
TDD: 0.75 u/kg 2/3 breakfast 1/3 before dinner Each dose should include: ¼ NPH ¼ Regular insulin	TDD: 1 u/kg 2/3 before breakfast 1/3 before dinner Each dose should include: 2/3 NPH 1/3 regular insulin

5.6.4.4. Discharge criteria

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5.6.4.4.1. Insulin dose is adjusted.

5.6.4.4.2. The family is taught about the insulin types, how to mix and inject the insulin, and how to rotate sites.

5.6.4.4.3. Health education discussing:

5.6.4.4.3.1. Short term complication e.g. (hypoglycemia, hyperglycemia and inter current illnesses and how to recognize and deal with it.

5.6.4.4.3.2. Home monitoring with glucometer" better with memory "initially before each meal time, before bed and once a week between 2 -4 AM.

5.6.4.4.3.3. To write readings in diary notebook which included at e and time of test.

5.6.4.4.3.4. Plan for meals and snacks and what must be done if RBS levels too high or low.

5.6.4.4.3.5. Long term complications

5.6.4.4.4. Written guidelines to the parents and school.

5.6.4.4.5. Ensure patient has necessary supplies and follow up appointment in diabetic clinic.

6.0 EQUIPMENTS

6.1. Glucometer

6.2. Gloves

6.3. Glucose stick

6.4. Lancet

6.5. IV set

6.6. Medications: Insulin, NaHCO3

6.7. IV Fluids

7.0 ATTACHMENT/FORMS

7.1. N/A

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8.0 REFERENCES

8.1. King Fahad Hospital, Policy and Procedure for Management of Diabetic Ketoacidosis, Policy Number: MDKA-PICU 3/3 (4)

9.0 REVISION CLAUSE

9.1. N/A

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